Rabbits and Myxomatosis

28

In 1859, Thomas Austin of Barwon Park Estate, Western Victoria, brought a consignment of rabbits to Australia from England and released them on his property for hunting purposes. They bred slowly at first but, after three years, they had become acclimatised and began to reach pest proportions. About this time a bushfire destroyed part of the fence around Austin's property. Rabbits escaped and spread dramatically.



Six years after their introduction, Austin killed 20 000 rabbits on his own property and estimated that there were at least another 10 000 still at large. By 1879 they were in New South Wales and, within a short time, they had spread over 75% of the continent, causing extensive damage to grazing land.

Rabbits had not been a serious pest in England despite their very high birth rate. This was because, in England, they had various 'natural enemies' which preyed on them and kept the population down. These natural enemies did not exist in Australia. Because of rabbit burrows and the habit of eating roots in dry seasons, the rabbit has destroyed areas which would otherwise have regenerated with the return of favourable rainfall. In this way, one rabbit can displace many sheep and cattle from a pasture.

Large sums of money were spent on trapping, poisoning, destruction of warrens and gassing but to no avail. In theory, one pair of rabbits, in ideal conditions, *could* multiply to 13 million rabbits in three years. Finally in the early 1950s, it was decided to try biological control. Specimens of the *myxoma* virus (originally found in South America) were injected into a few rabbits which were then released. The myxoma virus is carried by fleas and mosquitoes from live rabbits, and it spread very rapidly. By 1955, 90% of the rabbit population had died of the resulting disease, *myxomatosis*.

However, in recent years, two developments have taken place. Despite the fact that the virus is widespread, the rabbit population is increasing. The virus itself seems to be becoming less deadly.

Rabbits have varied in their resistance to myxomatosis. The disease did kill 90% of the population, leaving behind 10% which were those with the strongest resistance. The new population of rabbits in Australia is on its way to becoming a new species, compared to the English rabbit which could not resist myxomatosis. The myxoma virus is spread by fleas and mosquitoes from live rabbits, not from dead ones. The more virulent strains of the virus kill rabbits more quickly and, therefore, have less chance of surviving by spreading from one live rabbit to another. This helps to explain why the virus is becoming less virulent.

Our experience with rabbits and the myxoma virus supports the theory for natural selection. Those more fit (that is more resistant to the virus) survive, those less fit, do not. Evolution pushes species to change, when there is a change in circumstances.

1. From what overseas country did all the rabbits in Australia originate?
2. Why were they brought to Australia?
3. Why were they a serious pest in Australia when they were not in their original country?
4. Fencing was tried as one way to limit the spread of the rabbit population. Why do you think that this was not entirely successful?
5. Why was the myxoma virus introduced into the rabbit population?
6. What is the difference between the rabbit population of today and that pre-1950?
7. What does 'highly virulent' mean?
8. Does the experience with rabbits and myxomatosis support the theory of natural selection? Explain your answer.
9. Suggest a method of controlling the rabbit population that has not been mentioned already on this sheet. List advantages and disadvantages.

© Peter Gribben, Wendy St Germain and Emerald City Books, 2001. This sheet may be photocopied for non-commercial classroom use.
© Blake Education 978 1 92561 060 4 Assignments in Genetics Book 1